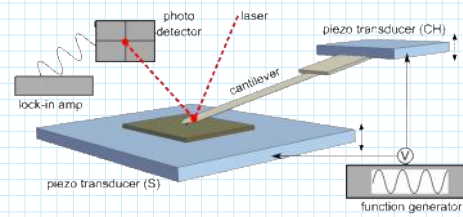


Atomic Force Microscopy

Thursday, May 22, 2025 2:14 PM



Scaling ~ 240nm
20.7
- 17.35

OSC AFM - Bruker

System
Username: nanoscope
Password: AFM2015

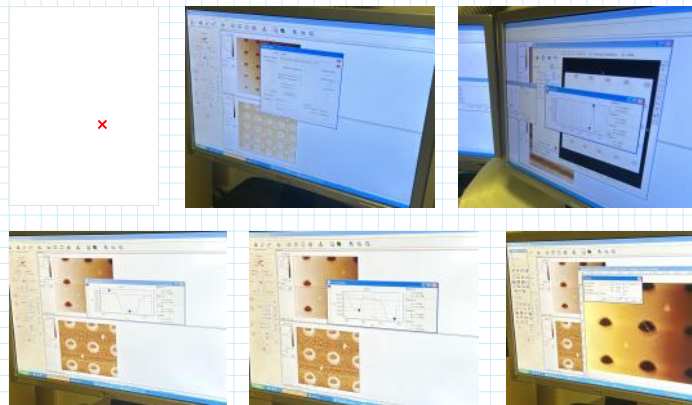


Calibration:

1. Use the calibration grid/gratings on the coin like mount
2. Use "Nanodrive" application
3. Figure the correct calibration file.
4. Run calibration for X, Y and Z (Z won't be aligned check again)
5. Use 'G' as the calibration grid (with holes)



https://email.arizona-my.sharepoint.com/personal/mohanbabumani_arizona_edu/Documents/RESEARCH-EICHENFIELD%20LAB/NOTES/EICHENFIELD%20LAB%20-%20Linked%20Files/AFM%20manual.pdf



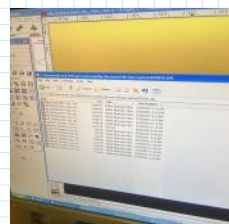
Measurement:

1. Place sample on the coin like mount with carbon tape
2. Place the coin into the AFM slot, close the apparatus and focus (find the sample and tip)
3. Use Nanodrive application to bring the tip to adjust the height by pressing 'auto engage'
4. Set range, sampling rate and speed (do not push beyond the threshold)
5. Start scanning
6. After the scan is complete the capture are stores in the PC folder
7. Use Gwyddion application to get rid of the streaks and other slopes caused by the variations and then calculate the RMS roughness of the sample measure (three level point measurement).
8. Use the scaling factor to adjust the 'dz' deviation caused by miscalibration of the tip (do the math to convert to RMS).

Speed:
0.1 Hz
20 μm/sec

Donot increase more than this!

Always keep track of the tip (parameters) used, calibration errors, scaling factors, formulas for getting the right RMS.



Park AFM at KECK center

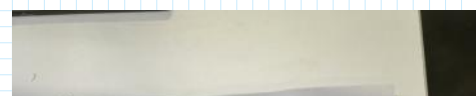
My door code : 615672 [Applies only for me]



Park SOP
Final

<https://arizona.app.box.com/s/fb10xe9c7odmjsqxl4wlnhnmssqsl45>

Video training

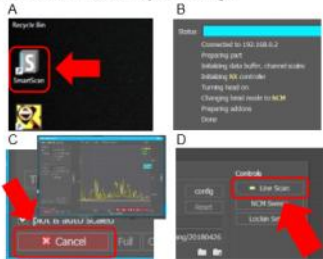


Park SOP – Updated 9/20/19

Keck Park NX20 AFM Standard Operation Procedure



1. Begin by:
- a. Opening SmartScan™ located on the computer desktop
 - b. Wait for the system to connect to the AFM
 - c. Cancel the "NCM Frequency Sweep" when the window opens, and close the window
 - d. If activated, disable "Line Scan" by deselecting it

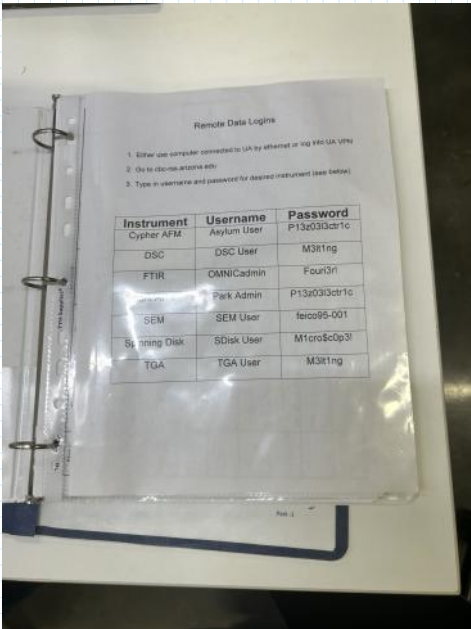


2. To enter the AFM setup guide:
- a. This can be done either in the Auto or Manual tab at the top of the screen. Choose either.
 - b. Left click on setup, located on the left of the screen and select "Probe", which will open an automated setup tutorial



3. The next series of steps will walk through loading the AFM cantilever and then sample loading. There will be animations helping to demonstrate the necessary things to do in each step.

Park -1



- a. Click Next on the AFM software. Watch the animation, note how the animation is instructing you on the following steps and showing how each one is done:



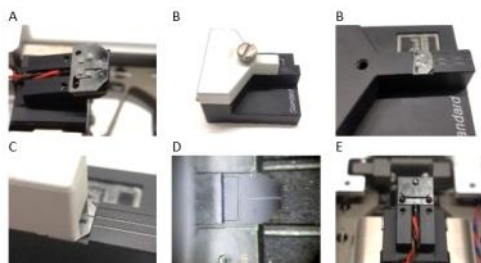
- i. Begin by opening the acoustic enclosure to access the AFM scanner and sample bed.
- ii. Turn off the laser if it is not already.
- iii. Fold the 2 butterfly clamps inward toward each other.
- iv. Slide the grey AFM scan head out and to the right. Do not touch the objective lens and hold the scanner carefully. It's very expensive (\$10K), so don't drop it!
- v. Hold the scanner on the back body, on the 2 rails that it slides on, then rotate it so that it is upside down.
- vi. Place the scanner in the clear plastic cradle on the desk, away from the edge of the table.



4. Next install an AFM cantilever. Make sure to click next after you removed the scan head and placed it on the cradle. The animation is lacking in this section. The important thing will be to remove the magnetic cantilever holder. This can be done with your fingers.

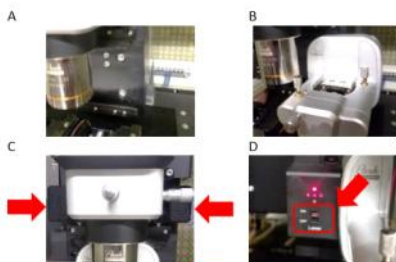


- a. Using your fingers gently hold the metal holder on either side and lift it up. The magnet is strong, so be ready for that. One trick is to move the metal chip parallel to the magnetic field until it slides out of it.
- b. Next place the probe holder into the cantilever loader. This unscrews and folds open. Place the probe holder so that it is centered onto the little metal stub.
- c. Tighten the knob until it is snug. Notice when the knob is tightened, there is a little flap that lifts. The cantilever substrate will slide into here.
- d. Slide a cantilever substrate under the flap. Note my picture and the approximate distance it is sticking out. **It is important to install all cantilevers about this distance.** If using an HQ300, align the middle etch line ~halfway between the flap and the outer edge. Once installed, loosen the knob on the holder to lower the tab and to secure the cantilever.
- e. Remove the metal holder and place it back onto the AFM scanner head. Note the 2 red dots at the bottom, the holder will rest on these.





5. Reinstall the AFM scan head back into the AFM.
 - a. Align the scanner with the rails, make sure that it is parallel to the ground.
 - b. Slide in, and gently wiggle it as you push, if it gets stuck, don't force it. Back it up and realign it.
 - c. Once it is connected, re-clamp the butterfly clamps so that the AFM scanner is secure.
 - d. Turn on the laser.

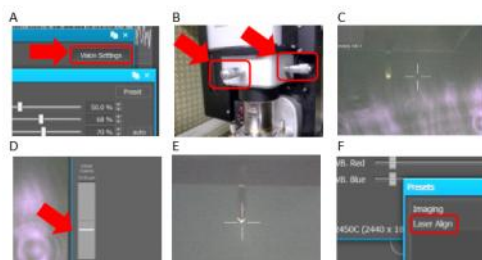


6. Once the head is reinstalled, always choose **yes** to cantilever type changed. Experiments use different types of cantilevers, and so never assume the previous user used the same one as you. Then select the cantilever that you are currently using. The basic one that we sell for general purpose imaging is an HQ300.

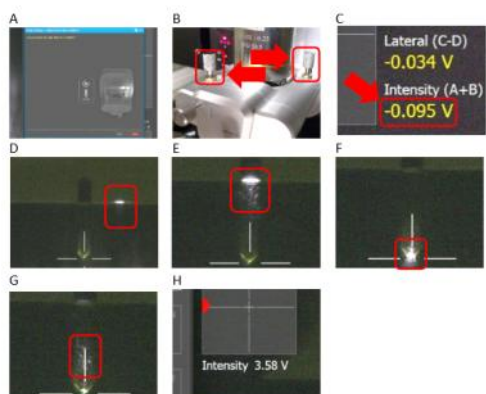


7. Next you must find the cantilever on the video screen, align it to the center and adjust the focus so that it is sharp. Align the cantilever to the middle.
 - a. To visualize the cantilever first click on the "video settings", then turn up the brightness, exposure and turn off auto contrast and set it to ~60.
 - b. To physically see the cantilever, identify the knobs on the microscope objective lens that will move the camera location.

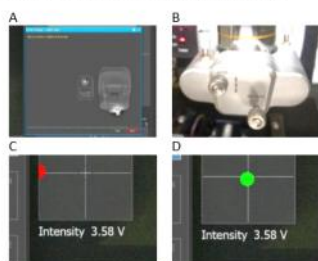
- c. Using these knobs, adjust the camera so that the cantilever is in the middle of the crosshairs on the video screen. Note that most the time, the forward backward knob is good for this.
- d. Use the focus control to then focus onto the cantilever so that the bottom edge is crisp and sharp. This is a click and hold control. To focus up, click above the centerline and hold. The farther you click above the centerline, the faster the focus will shift upward.
- e. Adjust the focus so that the cantilever is sharp, and centered on the video screen.
- f. Once the cantilever is sharp and clear, the video settings need to be changed before the next laser align step. This is accomplished by selecting the vision settings, and then choosing preset, and choosing "laser align". Select ok, close the settings, and select next to go to the next laser align AFM step.



8. To accurately measure surface topography with AFM, the laser must be aligned onto the back of the cantilever above where the AFM tip is.
 - a. After adjusting the video settings for laser align
 - b. The 2 control knobs on the top of the AFM scan head can be used to position the laser on the cantilever. Start by using the left knob. This moves the laser up and down, and typically is all that is needed to initially see the laser.
 - c. Walk the laser so that it is aligned at the top and center of the cantilever.
 - d. As you are adjusting the settings, if you don't see any changes to the laser intensity, turn off the laser and turn it back on. (The laser needs to be power cycled if you see -0.095 V)
 - e. Get the laser (which is a purple spot) to center onto the edge of the cantilever. It will be a bright spot! Make sure to align the real laser spot, and not a reflection onto the edge of the cantilever.
 - f. Walk the laser spot down so that the tip of the cantilever is just starting to illuminate with the laser.
 - g. Move the laser up, just to where the cantilever tip is. Adjust it left and right only to maximize the intensity signal.
 - h. Make sure the intensity is greater than 2.0. If it is not, then the laser is not correctly aligned.



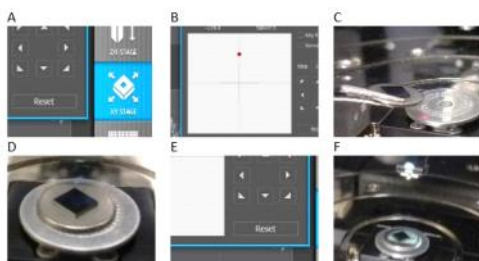
9. The last step in the alignment phase of imaging is the adjustment of the position-sensitive photo diode (PSPD).
- On the alignments page, nothing else is selected
 - Using the front 2 knobs adjust the PSPD so that, the signal is in the middle of the detector.
 - The red dot will be moved to the center of the crosshairs. Take your time, and the knobs will be turned a significant amount and then be very sensitive.
 - The intensity might increase once aligned, and the green dot doesn't have to be perfectly centered, just if its near the middle and green.



10. The final step in the setup algorithm is to select finish. Now that the system is on and aligned, the scanner can be moved in the z direction as well as the xy.



11. Loading a sample into the AFM is done with the XY bed pulled forward. The bed can only be moved once a cantilever is installed, and the PSPD has a green signal.
 - a. To move the bed, select the XY scanner in the lower right of the screen.
 - b. This is a variable click and hold control, much like the focus except in 4-directions. Click in the top center, to pull the scan bed forward and out.
 - c. The magnetic holder is extremely strong, bring the sample low, and to the side and slide into place.
 - d. Center your sample to be centered on the magnetic holder.
 - e. Select reset to have the system re-center the XY scanner below the AFM probe.
 - f. Visually verify this before continuing. If it is not, then adjust it using the 4-way direction control.



12. Next will be to select the imaging mode to begin imaging in.
 - a. This is done by selecting the manual tab on the top.
 - b. Then on the far left of the screen select the mode that you want. If using Tapping or NCM, a frequency sweep will be initiated.

- c. Make sure that the auto algorithm selects a peak near the cantilever harmonic frequency specified on the manufactures box. If it does not, you may have to select the frequency yourself.



13. Once the mode is selected, then the correct data channels to save must be chosen.

- a. The default for Tapping and NCM are show below. If doing a different mode, then they must be changed! To do this select the settings button on the top right of the channel menu on the bottom middle of the screen.
- b. Select the desired channels and hit apply to accept them. You can narrow them by selecting "Show All" and choose the image mode you want. Example for Non-contact mode and Tapping mode would be NCM Amplitude, NCM Phase, Z – Height, NCM Error.
- c. To choose where the data is saved and the file name to save, make sure to fill out file name, and select the target directory. All data must be saved in: Documents\Park Research Data\Your folder.



14. Now it's time to approach the sample and begin imaging. Select the menu item next to approach, verify that approach mode is on "Quick and Safe". To approach the surface, this is done with the approach button.

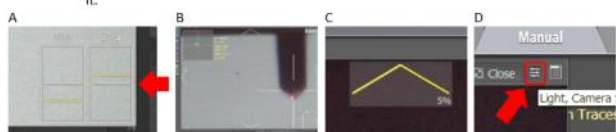


15. Once the system touches down onto the surface, we need to find the area that you want to image. This is done by moving the sample relative to the cantilever. To move the sample, you MUST lift the cantilever off the surface. Failure to do so WILL result in a broken tip, and you can restart by reloading a new cantilever. To lift, simply select the lift 100µm button. Alternatively, if you don't have to move far, you can double click within the video screen. If you are touched down, the system will auto-lift 100 µm and center the video screen where you clicked.

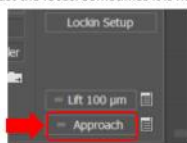


16. To find your sample or area of interest, you will need to focus onto the surface

- Do this with the focus control on the video screen. (The right control) When you lift up, you will need to focus up, and when you touch down, you will need to focus down. Be careful not to touch the Z-Height controls. Focus on your sample so that you can see features.
- You can zoom! Use the scroll wheel, and place the mouse on the video screen, wherever the mouse is, the zoom will center there.
- Moving the sample is done with arrows in the far-right edges of the video screen. To move, click, or click and hold. To adjust the movement speed, with the mouse hovered over any of the controls, use the scroll wheel to make the speed faster or slower. Moving can also be done by double clicking on the video screen. Where you double click, the sample bed will re-center to the center of the video screen.
- Sometimes you need to adjust the brightness and contrast. I like to set brightness at 100%, exposure at 7% and disable auto contrast and set the contrast to ~70. Feel free to choose your own settings, but this works well to enable the viewing of the sample, and reduce the laser spot bleed. There is a preset for these settings if you choose to use it.



17. Set image area and pixel density. After you find the area of interest, place the cantilever near where you want to image (within 100μm, because once you approach the surface, you only can move within a 100μm box before having to pickup and reposition the cantilever. Once satisfied, click approach and then adjust the focus. Sometimes it is necessary to adjust the brightness and

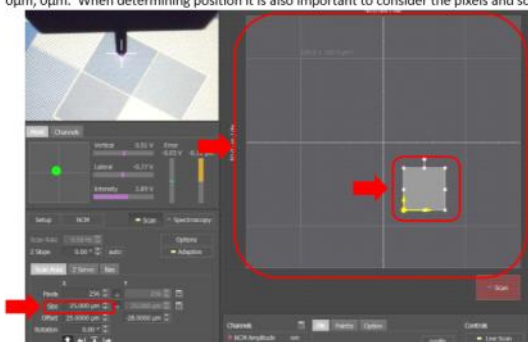


contrast.

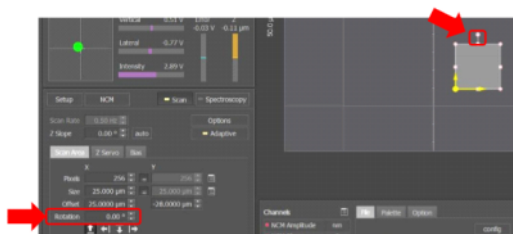
18. Once touched down on the surface, if line scan is not enabled, enable it. This will initiate the AFM to begin scanning back and forth (forward and reverse over a single area). It will continue to do so indefinitely until you tell it to scan or stop.
19. To know that you are on the surface and it engaged correctly, look to the trace-retrace for the height channel. If there are features on the trace retrace, then you are near the surface. If it looks like a noisy back and forth line, then you might have false engaged. Check the focus of the cantilever and the surface to make sure that both are in-focus or close to at the same plane.
 - a. Look at the Height channel, if you see features, then you are on the surface.
 - b. If you have doubts you are on the surface, increase the scan size. The default is 5 μ m.
 - c. At 25 μ m you should see some sort of feature if the sample is not atomically flat, and you are touched down in the height channel.



20. The position of the scan area can be adjusted within a 100 μ m scan area. This can be done by either manually entering offset values or dragging the scan box around. You will only see changes to the scan area if line-scan is enabled. To adjust the offset, on the scan area window, use the mouse scroll wheel to zoom out. Zoom until the outer boundary box for the 100 μ m box is displayed. Using the mouse, click hold and drag the scan area around until positioned. You cannot go outside the 100 μ m box. If you need to, then lift the cantilever, physically reposition the cantilever, and retouch down. Make sure to RESET THE OFFSET before touching down to 0 μ m, 0 μ m. When determining position it is also important to consider the pixels and scan size.



21. If you are imaging things for step height measurements, or film thicknesses, it is important to scan perpendicular to the edge of interest. To rotate the scan box, select the rotate too, and match the angle of the scan direction to be perpendicular. You can also enter an angle as well.



22. Now it is time to adjust the trace-retrace both the setpoint and the gain (how fast the system response to features) so that we can describe all the surface features in the scan box when we pass over them. We will be adjusting the gain, setpoint and scan speed.
- The gain is how fast the system responds to features. If the gain is too high, then noise will be introduced to the image, and it will be difficult to distinguish between noise and features. It is a fine line that is walked because the system must be sensitive enough to see all the surface features.
 - To adjust the gain, click on advanced, and start by adjusting the "I-gain" (integral gain). Slowly turn it up going in 0.25 increments and hitting the enter key after each adjustment. Note that the arrow up and down buttons suck, and you will have to manually type in and adjust the value. Sometimes it is necessary to have a I-gain that is less than 1. That is ok, remember the goal is to have the best trace-retrace as possible without adding noise. To know that it is good, watch the height channel to see if all the features are being described. Be careful to not turn the gain too high.
 - The best place to look and see if the gain is too high, is to look at the amplitude channel trace-retrace. You want to see each direction having a mirror/ reciprocal signal for each other. If you see a periodic oscillation in the channel, the gain is too high, and should be turned down.
 - If you are doing a step high measurement such as film thicknesses or chip heights, the +Z and -Z gain can be turned up. My recommendation is set these at 4, turning them up or down as needed. Then adjust the I-gain as needed.
 - The P-gain (proportional gain) should be set to be 1.5x the I-gain.

Touch down showing how the settings need to be adjusted for better trace/retrace



Gain is adjusted to high. Notice the noise in the amplitude trace/retrace



- b. The setpoint determines how close to the surface the system will keep the cantilever. Initially the system sets the setpoint at half the free-air amplitude of the oscillation of the cantilever. We often need to lower the setpoint. The lower the setpoint, the harder the system will push the cantilever onto the surface, without touching. In NCM mode, this means the system will push the cantilever closer to the surface, without touching. In tapping mode, this means the system keeps the cantilever closer to the surface and it is tapping off of it, so as you lower the setpoint you run the risk of breaking the cantilever.

- After the gain is set, and the image quality still is not improved, or some features are not being described, lower the setpoint slightly. Adjust it in 0.25nm increments hitting enter after each change. Push hard enough to describe all the features, but not too hard, that you run the risk of damaging the sample, or the cantilever. Watch the setpoint to make sure the features of interest are being described.
- Once the setpoint is lowered, often the gain needs to be adjusted and lowered.

Increasing the setpoint results in less noise, but poor trace/retrace without gain adjustment

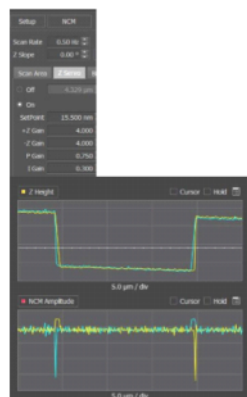


- c. After setting the setpoint and gain, adjust the scan speed for the image. If it is difficult to get a good trace-retrace, then slow down the image scan speed. The default is 1Hz,

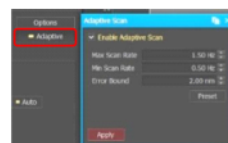
but it can be turned up or slowed down. One nice feature is the system has an adaptive scan rate, and can vary the speed based on the quality of the trace retrace.

- i. To enable adaptive, select advance, adaptive, and check the enabled box. I like to set the max rate at 1.5, the min at 0.5Hz. The error is your preference. The smaller the error, the system will stay slow, the larger, the faster the system will go. An error of 2nm is a good place.

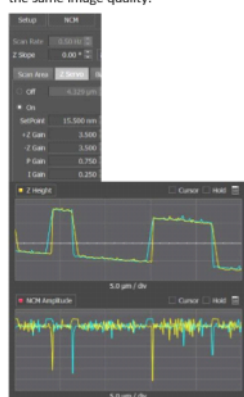
The scan speed is 0.5 Hz



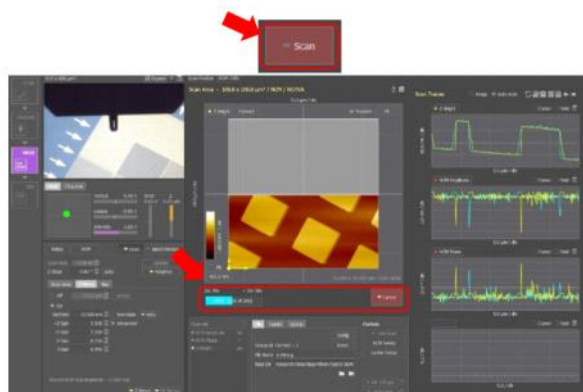
Enable adaptive through the advance menu



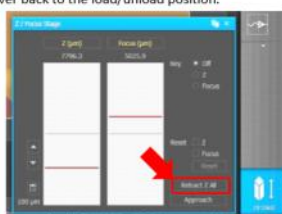
With adaptive it allows for faster scan times than a constant scan speed with the same image quality.



23. Once the gain, setpoint and speed are set, you can click scan! Note that every time you click scan, the system will scan over your area of interest and save each data channel to its own file. When and if you press cancel, a partial image will be saved.



24. After The scan is finished, it will begin line scanning at the opposite end of the scan area. Once the image is collected, it is dealer's choice on the next steps of, rescanning the current area, moving within the 100 μ m touchdown area and scanning again, lifting and then moving to a new area to scan, and lastly to lift, retract and shut the system down. The gain, setpoint and scan rate and size will need to be set for every image taken.
25. To end the imaging session, lift the cantilever off the surface, then select "Retract Z All". This will lift the cantilever back to the load/unload position.



26. Next pull the sample forward so that you can access it for unloading using the XY stage control. Remember the center top moves the sample towards the front of the AFM enclosure.



27. To shutdown, close the software:
 - a. Remove the sample
 - b. Turn off the laser, unclamp the scan head
 - c. Remove the AFM scan head
 - d. Remove the cantilever, then return the cantilever holder to the scan head
 - e. Reinstall the scan head onto the AFM
 - f. Clamp in place, leave with laser off.
 - g. Replace/turnoff any equipment turned on or removed for sample securement
 - h. Close the enclosure
28. Sign both the digital and paper logbook and report the materials used. Congratulations you got your data and are now an AFM expert. Happy imaging!